Estimate the size of answers

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: ________________

1. Read the model: Estimate 6,782 x 49. Copy or complete the first step.

2. Estimate 92,410 / 31.

3. Estimate 487,205 + 218,770 - 96,420.

4. Miro says: Miro rounds every number to one significant digit even when this gives a poor estimate. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Estimate 6,782 x 49. Copy or complete the first step.
   Answer About 6,800 x 50 = 340,000.

2. Estimate 92,410 / 31.
   Answer About 90,000 / 30 = 3,000.

3. Estimate 487,205 + 218,770 - 96,420.
   Answer About 490,000 + 220,000 - 100,000 = 610,000.

4. Miro says: Miro rounds every number to one significant digit even when this gives a poor estimate. Is this correct? Explain.
   Answer Choose compatible rounded values that keep the calculation easy and reasonably close.

Estimate the size of answers

Work independently. Show a complete method and check at least one answer.

Name: ______________________________ Date: ____________

1. Estimate 6,782 x 49.

2. Estimate 92,410 / 31.

3. Estimate 487,205 + 218,770 - 96,420.

4. Solve this independently and show every stage: Estimate 6,782 x 49.

5. Use a second method or representation for: Estimate 92,410 / 31.

6. Check this result using an inverse, estimate or known fact: Estimate 487,205 + 218,770 - 96,420.

7. A calculator shows 7,204 x 58 = 41,783. Explain without exact calculation why it is wrong.

8. Identify and correct this misconception: Miro rounds every number to one significant digit even when this gives a poor estimate.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Estimate 487,205 + 218,770 - 96,420.

Teacher answers and checking prompts

1. Estimate 6,782 x 49.
   Answer About 6,800 x 50 = 340,000.

2. Estimate 92,410 / 31.
   Answer About 90,000 / 30 = 3,000.

3. Estimate 487,205 + 218,770 - 96,420.
   Answer About 490,000 + 220,000 - 100,000 = 610,000.

4. Solve this independently and show every stage: Estimate 6,782 x 49.
   Answer About 6,800 x 50 = 340,000.

5. Use a second method or representation for: Estimate 92,410 / 31.
   Answer Expected result: About 90,000 / 30 = 3,000. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Estimate 487,205 + 218,770 - 96,420.
   Answer Expected result: About 490,000 + 220,000 - 100,000 = 610,000. A valid check must be shown.

7. A calculator shows 7,204 x 58 = 41,783. Explain without exact calculation why it is wrong.
   Answer The product should be about 7,200 x 60 = 432,000, so the displayed answer is far too small.

8. Identify and correct this misconception: Miro rounds every number to one significant digit even when this gives a poor estimate.
   Answer Choose compatible rounded values that keep the calculation easy and reasonably close.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Estimate 487,205 + 218,770 - 96,420.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Estimate the size of answers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ____________

1. A calculator shows 7,204 x 58 = 41,783. Explain without exact calculation why it is wrong.

2. Prove the answer to this question, rather than only calculating it: Estimate 6,782 x 49.

3. Solve this in two different ways and compare them: Estimate 92,410 / 31.

4. Work backwards from the answer to reconstruct the question: About 490,000 + 220,000 - 100,000 = 610,000.

5. Explain why this reasoning fails: Miro rounds every number to one significant digit even when this gives a poor estimate.

6. Create a new example that tests the same idea as: Estimate 487,205 + 218,770 - 96,420.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. A calculator shows 7,204 x 58 = 41,783. Explain without exact calculation why it is wrong.
   Answer The product should be about 7,200 x 60 = 432,000, so the displayed answer is far too small.

2. Prove the answer to this question, rather than only calculating it: Estimate 6,782 x 49.
   Answer Expected result: About 6,800 x 50 = 340,000. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Estimate 92,410 / 31.
   Answer Expected result: About 90,000 / 30 = 3,000. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: About 490,000 + 220,000 - 100,000 = 610,000.
   Answer One valid reconstruction is: Estimate 487,205 + 218,770 - 96,420. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro rounds every number to one significant digit even when this gives a poor estimate.
   Answer Choose compatible rounded values that keep the calculation easy and reasonably close.

6. Create a new example that tests the same idea as: Estimate 487,205 + 218,770 - 96,420.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.